

大学物理课后练习十二答案

一、选择题

1. (D)

解: 如图 1-1, 以 O 为原点 r 为半径作高斯面 S , 由高斯定理 $\int \vec{E} \cdot d\vec{S} = 0$, $E = 0$, ($r < R_1$)

2. (B)

解: 如图 1-2, 由电势叠加原理, P 点电势是由电荷 Q 和 q 分别在 P 点产生的电势的叠加

$$U_P = U_1 + U_2 = \frac{Q}{4\pi\epsilon_0 R} + \frac{q}{4\pi\epsilon_0 r} = \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{R} + \frac{q}{r} \right)$$

3. (A)

解: 如图 1-3, 在两球面间作高斯面, 得 $E = \frac{q}{4\pi\epsilon_0 r^2}$, 得 $U_1 - U_2 = \int \vec{E} \cdot d\vec{r} = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{1}{R} \right)$

4. (A)

解: 均匀带电物体面内各处电势相等, 为 $\frac{Q}{4\pi\epsilon_0 R}$, 球面外电势为 $\frac{Q}{4\pi\epsilon_0 r}$, 与 r 成反比。

5. (D)

解: 由 $\vec{E} = -\frac{dV}{dl_n} \vec{e}_n$, 当 $\frac{dV}{dl_n} < 0$ 时, $E > 0$, 即 $\vec{E}$ 的方向总是由高电势处指向低电势处。

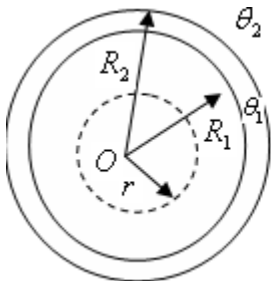


图 1-1

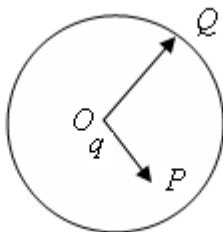


图 1-2

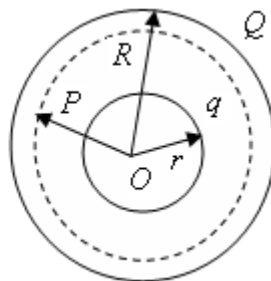


图 1-3

二、填空题

1. $\pm \frac{\sigma}{2\epsilon_0} \pi R^2$

解: 场强大小为 $\frac{\sigma}{2\epsilon_0}$, 方向沿平面法线方向, 故 $\phi_e = \int_s E \cos \theta ds = \pm \frac{\sigma}{2\epsilon_0} \pi R^2$

2. $E_A = \frac{\sigma}{2\epsilon_0}$; $E_B = \frac{3\sigma}{2\epsilon_0}$; $E_C = -\frac{\sigma}{2\epsilon_0}$

解: 场强叠加: $E_A = -\frac{\sigma}{2\epsilon_0} + \frac{2\sigma}{2\epsilon_0} = \frac{\sigma}{2\epsilon_0}$, $E_B = \frac{\sigma}{2\epsilon_0} + \frac{2\sigma}{2\epsilon_0} = \frac{3\sigma}{2\epsilon_0}$, $E_C = \frac{\sigma}{2\epsilon_0} - \frac{2\sigma}{2\epsilon_0} = -\frac{\sigma}{2\epsilon_0}$

$$3. \frac{q}{6\pi\epsilon_0 L}; \quad \frac{q}{6\pi\epsilon_0 L}$$

解: $U_0 = \frac{q}{4\pi\epsilon_0 L} + \frac{-q}{4\pi\epsilon_0 L} = 0$, $U_D = \frac{q}{4\pi\epsilon_0 3L} + \frac{-q}{4\pi\epsilon_0 L} = -\frac{q}{6\pi\epsilon_0 L}$, $U_{OD} = \frac{q}{6\pi\epsilon_0 L}$, 因此

$$W_{OD} = \frac{q \times 1}{6\pi\epsilon_0 L}, \text{ 又因为 } U_\infty = 0, \text{ 故 } W_{D\infty} = \frac{q}{6\pi\epsilon_0 L}$$

$$4. \frac{q_1 q_2}{4\pi\epsilon_0} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

$$5. \frac{Qq}{4\pi\epsilon_0 r}$$

$$6. \sigma = \frac{U\epsilon_0}{r_1 + r_2}; \quad 4\pi r_2^2 \sigma \left(1 + \frac{r_1}{r_2} \right) \text{ 或 } 4\pi\epsilon_0 r_2 U$$

$$7. E_x = \frac{-2Bx}{x^2 + y^2}; \quad E_z = 0$$

解: 由 $E_x = -\frac{\partial U}{\partial x} = -\frac{\partial}{\partial x} B \ln(x^2 + y^2) = \frac{-2Bx}{x^2 + y^2}$, $E_z = -\frac{\partial U}{\partial z} = -\frac{\partial}{\partial z} B \ln(x^2 + y^2) = 0$

$$8. \frac{q}{8\pi\epsilon_0 l} \ln \frac{x+2l}{x}; \quad \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{x(x+2l)}$$

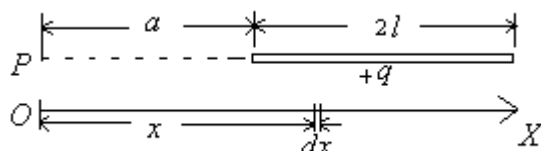
解: (1) 以 p 点为坐标原点建立坐标系, 方向如图所示

$$dq = \lambda dx, \quad \lambda = \frac{q}{2l}, \quad dV = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda dx}{x}$$

$$V = \int_l dV = \frac{\lambda}{4\pi\epsilon_0} \int_x^{x+2l} \frac{dx}{x} = \frac{\lambda}{4\pi\epsilon_0} \ln \frac{x+2l}{x} = \frac{q}{8\pi\epsilon_0 l} \ln \frac{x+2l}{x}$$

(2) 以 x 为变量, 由电场强度与电势差的关系可得:

$$E = -\frac{\partial V}{\partial a} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{x(x+2l)}$$



三、计算题

1. 解：以 O 为球心做半径为 r 的球形高斯面

$$(1) \quad r < R_1 \text{ 时, 由高斯定理 } \int \vec{E}_1 \cdot d\vec{S} = E_1 \cdot 4\pi r^2 = 0, \text{ 得 } E_1 = 0$$

$$(2) \quad R_1 < r < R_2 \text{ 时, } \int \vec{E}_2 \cdot d\vec{S} = E_2 \cdot 4\pi r^2 = \frac{1}{\varepsilon_0} \rho \left(\frac{4}{3} \pi r^3 - \frac{4}{3} \pi R_1^3 \right), \text{ 得 } E_2 = \frac{\rho}{3\varepsilon_0} \cdot \frac{r^3 - R_1^3}{r^2}$$

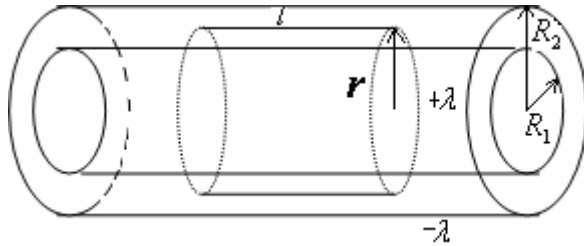
$$(3) \quad r > R_2 \text{ 时, } \int \vec{E}_3 \cdot d\vec{S} = E_3 \cdot 4\pi r^2 = \frac{1}{\varepsilon_0} \rho \left(\frac{4}{3} \pi R_2^3 - \frac{4}{3} \pi R_1^3 \right), \text{ 得 } E_3 = \frac{\rho}{3\varepsilon_0} \cdot \frac{R_2^3 - R_1^3}{r^2}$$

2. 解：作半径为 r ，高为 l 的同轴圆柱面加上、下底面为高斯面，如图

$$(1) \quad r < R_1 \text{ 时, } \oint_S \vec{E}_1 \cdot d\vec{S} = E_1 \cdot 2\pi r l = 0, \text{ 得 } E_1 = 0$$

$$R_1 < r < R_2 \text{ 时, } \oint_S \vec{E}_2 \cdot d\vec{S} = E_2 \cdot 2\pi r l = \frac{\lambda l}{\varepsilon_0}, \text{ 得 } E_2 = \frac{\lambda}{2\pi\varepsilon_0 r}, \text{ 方向沿径向向外。}$$

$$r > R_2 \text{ 时, } \oint_S \vec{E}_3 \cdot d\vec{S} = E_3 \cdot 2\pi r l = 0, \text{ 得 } E_3 = 0$$



$$(2) \quad \Delta V = \int_{R_1}^{R_2} \vec{E} \cdot d\vec{S} = \int_{R_1}^{R_2} \frac{\lambda}{2\pi\varepsilon_0 r} dr = \frac{\lambda}{2\pi\varepsilon_0} \ln \frac{R_2}{R_1}$$

$$(3) \quad r < R_1 \text{ 时, } V = \int_r^{R_2} \vec{E} \cdot d\vec{l} = \int_{R_1}^{R_2} \vec{E} \cdot d\vec{l} = \int_{R_1}^{R_2} \frac{\lambda}{2\pi\varepsilon_0 r} dr = \frac{\lambda}{2\pi\varepsilon_0} \ln \frac{R_2}{R_1}$$

$$R_1 < r < R_2 \text{ 时, } V = \int_r^{R_2} \vec{E} \cdot d\vec{l} = \int_r^{R_2} \frac{\lambda}{2\pi\varepsilon_0 r} dr = \frac{\lambda}{2\pi\varepsilon_0} \ln \frac{R_2}{r}$$

$$r > R_2 \text{ 时, } V = \int_r^{R_2} \vec{E} \cdot d\vec{l} = 0$$

$$3. \text{ 解: } E = \frac{\lambda}{2\pi\varepsilon_0 r} = \frac{\lambda}{4\pi\varepsilon_0 a}$$

$$F = -\lambda E = -\frac{\lambda^2}{4\pi\varepsilon_0 a}$$

电场力垂直于带电直线且为吸引力

4. 解: (1) 取薄球壳 $r \rightarrow r + dr$, $dQ = \rho dV = \frac{qr}{\pi R^4} 4\pi r^2 dr$, $Q = \int dQ = \int \frac{4qr^3}{R^4} dr = q$

(2) 利用高斯定理: $r < R$ 时, $\int \vec{E}_1 \cdot d\vec{S} = E_1 \cdot 4\pi r^2 = \frac{1}{\varepsilon_0} \int_0^r \frac{4qr^3}{R^4} dr = \frac{qr^4}{\varepsilon_0 R^4}$, 得 $E_1 = \frac{qr^2}{4\pi\varepsilon_0 R^4}$

$r > R$ 时, $\int \vec{E} \cdot d\vec{S} = E_2 \cdot 4\pi r^2 = \frac{q}{\varepsilon_0}$, 得 $E_2 = \frac{q}{4\pi\varepsilon_0 r^2}$

(3) 选无穷远处为电势零点, $r < R$ 时,

$$V_1 = \int_r^R \vec{E} \cdot d\vec{r} + \int_R^\infty \vec{E}_2 \cdot d\vec{r} = \int_r^R \frac{qr^2}{4\pi\varepsilon_0 R^4} dr + \int_R^\infty \frac{q}{4\pi\varepsilon_0 r^2} dr = \frac{q}{12\pi\varepsilon_0 R^4} (4R^3 - r^3)$$

$$r > R \text{ 时, } V_2 = \int_r^\infty \vec{E}_2 \cdot d\vec{r} = \frac{q}{4\pi\varepsilon_0 r}$$